



澳門科技大學
MACAU UNIVERSITY OF SCIENCE AND TECHNOLOGY

BE202 EX4 Final Project Proposal

Glove Mouse

Xie, Wen-hao 1230021133

Shi, Zhi-bo 1230017906

Background

- The development of a wearable gesture-based glove mouse is essential for enhancing user interaction and accessibility in computing. This device enables cursor control through natural hand movements, offering a more intuitive and interesting experience.



Description

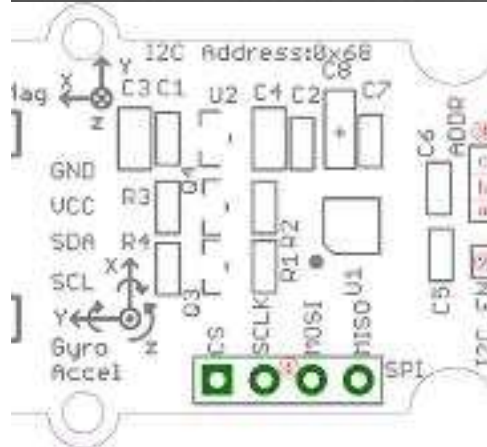
In this project, we will:

1. Use an accelerometer and gyroscope to detect hand movements and gestures for cursor control.
2. Implement a capacitive touch sensor to register finger touches for mouse clicks.
3. Process the sensor data with an Arduino, translating movements and touches into computer inputs.
4. Connect the Arduino to a computer via USB for seamless interaction.

Special Devices

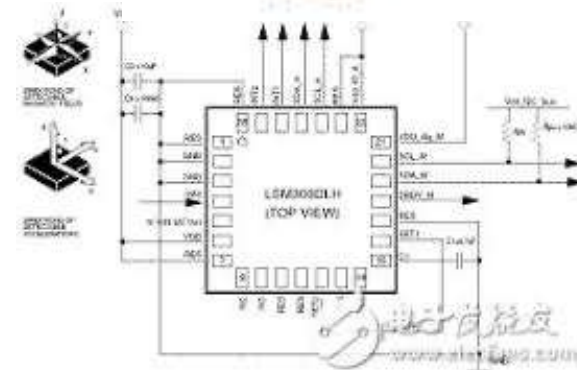
**9DOF (9 Degrees of Freedom)
breakout board**

LSM9DS0



**magnetometer and
accelerometer**

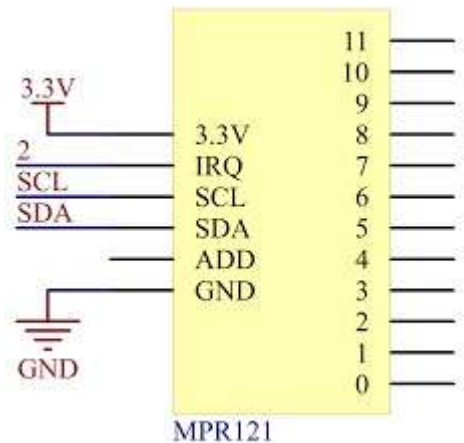
LSM303



Special Devices

Close capacitive touch sensor
controller

MPR121



accelerometer and gyroscope

MPU-6050

